

Structural Search

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Introduction

Overall plan

#	Topic		Content
1	Bertrand	T	
2	Demand Estimation I	E	OLS, AIDS, Logit
3	Demand Estimation II	E	Logit extensions, supply, mergers
4	Demand Estimation III	E	BLP
5	Price Dispersion	T	
6	Search: Homogenous	T	
7	Search: Heterogenous	T	
8	Structural Search	E	BLP, but ε_{jt} must be searched
9	Advertising	E	Firms choose <i>ads</i>
10	Digital Markets I	E	Applications
11	Digital Markets II	T	
12	Platforms I	E	Applications
13	Platforms II	T	
14	Privacy	T	

Plan for today

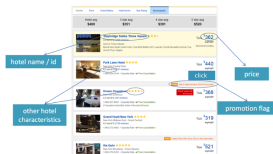
- BLP with Search – how?
- MSW: searching over *dealerships* of cars
- Ursu (2018): Source of the data for project 2

Figure 1: Information on the Data Observed

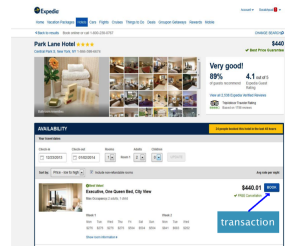


The image shows the Expedia search interface. At the top, there are input fields for 'visitor country, region', 'time of search', and 'site name'. Below these are navigation links: 'Home', 'Vacation Packages', 'Hotels', 'Cars', 'Flights', 'Cruises', and 'Things to do'. The main section is titled 'PLAN YOUR TRIP ON EXPEDIA' and contains a search form. The form has tabs for 'Flight', 'Hotel', 'Car', 'Activities', and 'Cruise'. The 'Hotel' tab is selected. Below the tabs, there are input fields for 'Find hotels near:' (with a dropdown for 'A city, airport or attraction'), 'What City?', 'length of stay' (with 'mm/dd/yyyy' and 'mm/dd/yyyy' fields), 'booking window' (with a dropdown), '# rooms', '# adults', '# children', and 'Room 1' (with '2' and '0' fields). There is a 'SEARCH FOR HOTELS' button and a 'BEST PRICE GUARANTEE' badge.

Search Query



Search Impression: Information Before the Click



The image shows the Expedia hotel page for 'Park Lane Hotel'. The page includes a header with navigation links, a main section with hotel photos and a 'Very good!' rating of 4.1 out of 5, and an 'AVAILABILITY' section. The 'AVAILABILITY' section shows a table with columns for 'Check-in', 'Check-out', 'Rooms', 'Adults', and 'Children'. The table has two rows: one for 'Check-in' and one for 'Check-out'. The 'Check-in' row shows '10/23/2018' and '10/23/2018'. The 'Check-out' row shows '10/24/2018' and '10/24/2018'. The 'Rooms' column shows '1' and '2'. The 'Adults' column shows '2' and '2'. The 'Children' column shows '0' and '0'. The 'Price' column shows '\$440.01' and '\$440.01'. A blue box labeled 'transaction' points to the '\$440.01' price.

Hotel's Page: Information After the Click

Let's play a game

Search with dice

1. Roll 10 dice under cups ("boxes") so you can't see the outcomes
2. Start by opening one box
3. Each additional box you open costs 1 point

Final score: the count on the best dice result you have seen.

Question: At what dice value should you stop?

Solving the dice game

- **Task:** Figure out the optimal *stopping rule*
 - I.e. at what dice outcome, z , should you stop opening cups?
- **First cup:** The expected value of this cup is

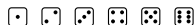
$$\mathbb{E}(X) = \sum_{k=1}^6 \frac{1}{6} k = \frac{1}{6} (1 + 2 + 3 + 4 + 5 + 6) = 3.5.$$

- **Second cup:** Suppose you rolled $X = 3$ at first. Gain from opening one more box:

$$\begin{aligned} \underbrace{\mathbb{E} [\max(X - 3, 0)]}_{=H(3)} &= \sum_{k=1}^6 \frac{1}{6} \max(k - 3, 0) \\ &= \frac{1}{6} (0 + 0 + 0 + 1 + 2 + 3) = 1. \end{aligned}$$

- **Conclusion:** At $z = 3$, the gain equals search cost ($c = 1$), so we are *indifferent* between searching and stopping.

- **Reservation values** by the current utility $H(z) \equiv \mathbb{E} \max\{u - z, 0\}$ (the state variable)



$$H(0) = \frac{1}{6} (1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} \cong 3.5$$

$$H(1) = \frac{1}{6} (0 + 1 + 2 + 3 + 4 + 5) = \frac{15}{6} \cong 3.5$$

$$H(2) = \frac{1}{6} (0 + 0 + 1 + 2 + 3 + 4) = \frac{15}{6} \cong 3.5$$

$$H(3) = \frac{1}{6} (0 + 0 + 0 + 1 + 2 + 3) = \frac{6}{6} = 1.0$$

$$H(4) = \frac{1}{6} (0 + 0 + 0 + 0 + 1 + 2) = \frac{3}{6} \cong 0.5$$

$$H(5) = \frac{1}{6} (0 + 0 + 0 + 0 + 0 + 1) = \frac{1}{6} \cong 0.2$$

$$H(6) = \frac{1}{6} (0 + 0 + 0 + 0 + 0 + 0) = \frac{0}{6} = 0$$

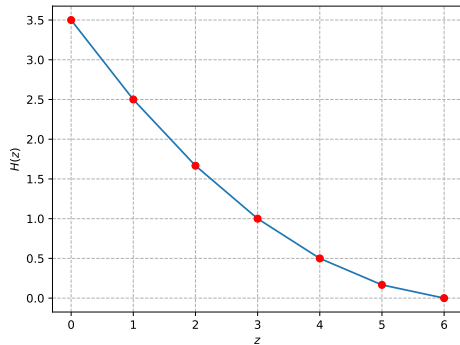
More graphically

- **Reservation values** are the row sums times $\frac{1}{6}$: $H(z) = \sum_{k=1}^6 \frac{1}{6} \max\{k - z, 0\}$

		dice outcome						
								$H(z)$
current value (z)	0	1	2	3	4	5	6	3.5
	1	0	1	2	3	4	5	2.5
	2	0	0	1	2	3	4	1.7
	3	0	0	0	1	2	3	1.0
	4	0	0	0	0	1	2	0.5
	5	0	0	0	0	0	1	0.2
	6	0	0	0	0	0	0	0.0

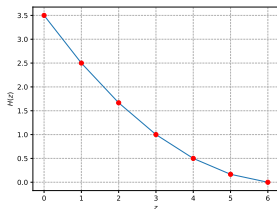
Visualizing $H(z)$

- **Value of search** is $H(z) \equiv \mathbb{E} \max\{u - z, 0\}$.



Visualizing $H(z)$

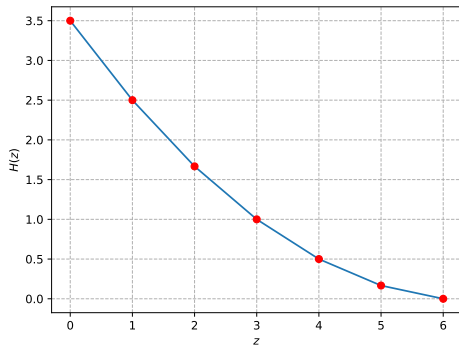
- **Value of search** is $H(z) \equiv \mathbb{E} \max\{u - z, 0\}$.



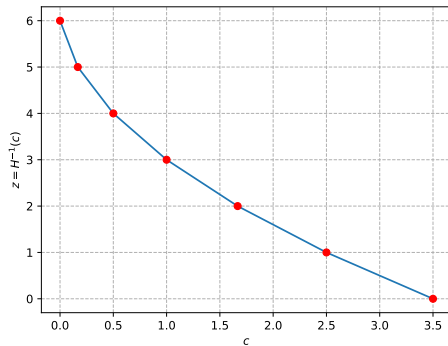
- **Optimal strategy:** Continue opening boxes until *reservation value* is met: $u \geq r \equiv H^{-1}(c)$.
- **Note:** Function is clearly **strictly monotonic** (and even **convex**) $\Rightarrow H^{-1}$ is well-defined!

Visualizing $z := H^{-1}(c)$

Value of search $H(z) \equiv \mathbb{E}[\max\{u - z, 0\}]$



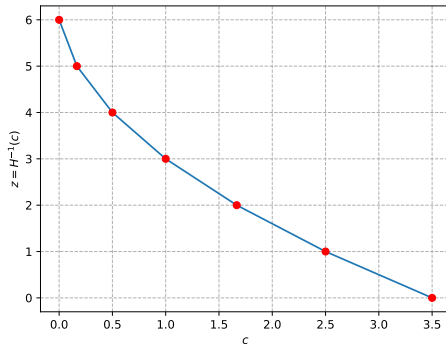
The “reservation value” $r \equiv H^{-1}(c)$



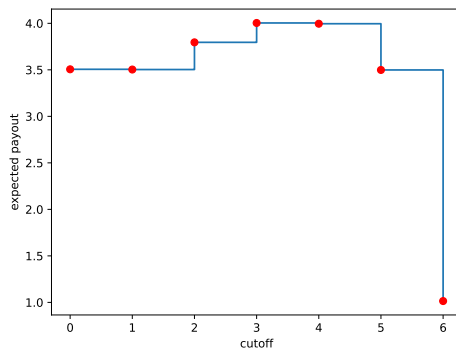
Reservation value is $r = H^{-1}(c)$

- **Rule:** stop searching if $u \geq r$ satisfying

$$H(r) = c \Leftrightarrow r = H^{-1}(c).$$



Performance of cutoff rules



Technical note: While $H(z)$ is continuous, the simulated payoff is a step function, since the rolled dice outcomes are integers, so setting cutoffs between integers makes no difference.

Model

- **Consumption utility:** $u_{ij} = \delta_{ij} + \varepsilon_{ij}$, where $\varepsilon_{ij} \sim \text{IID} F^\varepsilon$
- **Search costs:** $c_{ij} \sim F^c(\cdot | \mu_{ij})$
- **Search decisions** are to sequentially choose what to search.
- **Final purchase:** Choose best product amongst those searched.
- **Outside option:** Yields utility zero; sometimes we assume $c_{i0} = 0$.
- **Two models:** “MSW” and “Ursu”
 - Differ in what they assume for F^ε and F^c
 - Agree in setting $\delta_{ij} = \mathbf{x}_{ij}^{u'} \boldsymbol{\beta}$ and $\mu_{ij} = \mathbf{x}_{ij}^{s'} \boldsymbol{\gamma}$.
 - MSW is easy to estimate; Ursu is easy to simulate.

The two models you will meet

- **MSW:** Moraga-Gonzales, Sandor & Wildenbeest (2022; Restud): *"Consumer Search and Prices in the Automobile Market"*
- **Ursu:** Ursu (2018; Marketing Science): *"The Power of Rankings: Quantifying the Effect of Rankings on Online Consumer Search and Purchase Decisions"*

		Ursu	MSW
F^0	CDF for ε_{ij}	Gaussian	T1EV
F^c	CDF for c_{ij}	$F_{ij}^c(c) = \Phi\left(\frac{c - \mu_{ij}}{\sigma_c}\right)$	$F_{ij}^c(c) = \frac{1 - F[H_0^{-1}(c) + \mu_{ij}]}{1 - F[H_0^{-1}(c)]}$
	Simulation	Easy	Tricky (F^c is hard)
	$\Pr(i \text{ chooses } j)$	Complicated	Logit: $\frac{\exp(\delta_{ij} - \mu_{ij})}{\sum_{k \in \mathcal{J}} \exp(\delta_{ik} - \mu_{ik})}$

(technically, Ursu sets $c_{ij} = \exp(\mu_{ij})$, corresponding to $\sigma_c = 0$.)

- **The agent's problem:**

- Prior to search: The agent i knows (δ_{ij}, c_{ij}) for all j .
- Searching j reveals ε_{ij} .
- Problem: How to choose what to search and when to stop?

- Define the **Value of Search** when the agent already has found a value of z

$$H_{ij}(z) \equiv \mathbb{E}_{u_{ij}}[\max(u_{ij} - z, 0)]$$

- Define the **Reservation Value** r_{ij} as the point of indifference

$$H_{ij}(r_{ij}) = c_{ij}.$$

- **Recall Pandora's rule:** Open the box next with the highest r_{ij} , unless it is below the best utility encountered.
- **Armstrong (2017):** Define $w_{ij} = \min\{r_{ij}, u_{ij}\}$. The agent chooses $\arg \max_j w_{ij}$.

The Value of Search

Given a reference utility of $z \in \mathbb{R}$, the expected value of searching j is

$$H_{ij}(z) \equiv \mathbb{E} \max\{u_{ij} - z, 0\}.$$

- **Define** the function

$$H_0(z) \equiv \int_{-\infty}^{\infty} \max(\varepsilon - z, 0) dF_0^\varepsilon(\varepsilon).$$

- **By substituting** $u_{ij} = \delta_{ij} + \varepsilon_{ij}$, we see

$$H_{ij}(z) \equiv \mathbb{E}_{u_{ij}} [\max(u_{ij} - z, 0)] = \mathbb{E}_{\varepsilon_{ij}} [\max(\varepsilon_{ij} - (z - \delta_{ij}), 0)] \equiv H_0(z - \delta_{ij}).$$

- **Fact:** since F_0^ε is continuous and strictly monotonic, $H_0(\cdot)$ is strictly decreasing
 $\Rightarrow H_0$ has a well-defined inverse, H_0^{-1} , and

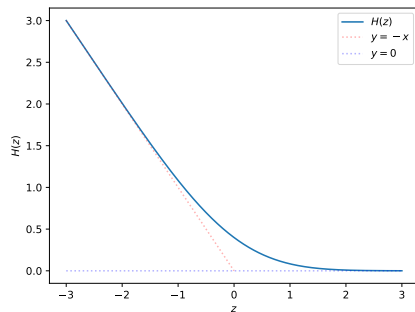
$$H_{ij}^{-1}(c) = \delta_{ij} + H_0^{-1}(c).$$

- **Analytical solution(!):** In the Gaussian case, $F^\varepsilon = \Phi$, you can show that

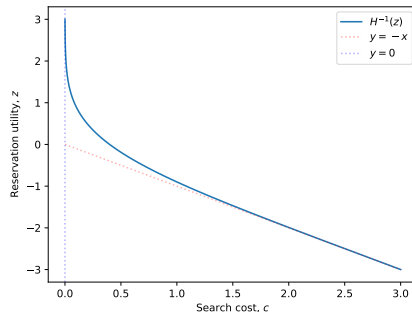
$$H_0(z) = \phi(z) - z [1 - \Phi(z)].$$

H_0 in the Gaussian Case ($F^\varepsilon = \Phi$)

Value of Search: $H_0(z)$



Reservation value: $H_0^{-1}(c)$



Discuss: Why does $H_0(z) \searrow 0$ as $z \rightarrow \infty$? And why is $H_0(z) > 0$ regardless of how large z is?

Simulating Data

Given $(\delta_{ij}, c_{ij}, \varepsilon_{ij})$, compute $u_{ij} = \delta_{ij} + \varepsilon_{ij}$ and pre-compute $H_0^{-1}(\cdot)$. Then do:

1. Compute reservation value $r_{ij} = H_0^{-1}(c_{ij})$
2. Compute Weitzman index $w_{ij} = \min(r_{ij}, u_{ij})$
3. Assign choice $d_i = \arg \max_{j \in \mathcal{J}} w_{ij}$.

- **Ursu:** $c_{ij} \sim \mathcal{N}(\exp(\mu_{ij}), \sigma_C^2)$ with $\sigma_C = 0$ (!).
- **Numerical challenge:** We will have to invert $H_0(z)$ *numerically*.
 - Compute $\{z, H_0(z)\}_{z \in \mathcal{Z}}$ for some grid $\mathcal{Z}$, and interpolate between the points.

- **Motivation:** If we want to estimate parameters, we will need Conditional Choice Probabilities (CCPs)...
- **In general**, since $d_i = \arg \max_j w_{ij}$, we have

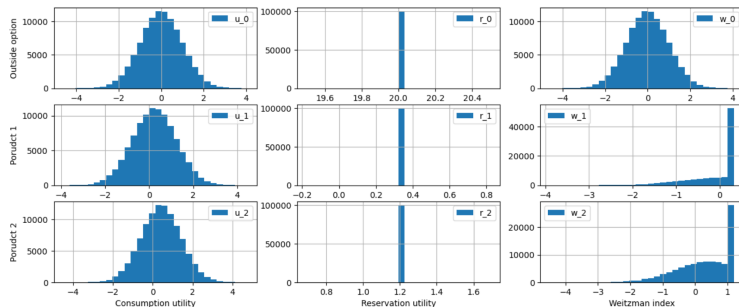
$$\Pr(i \text{ chooses } j) = \Pr(w_{ij} \geq w_{ik} \forall k).$$

- **Brute force simulation:** Just simulate the model many times and see how often it happens...
 - ... that just takes a long time.
- **MSW:** MSW instead ask “*what do we need to assume about F_{ij}^C so that $w_{ij} \sim T1EV?$ ”*

A Simulation Example

Simulated distributions of $(u_{ij}, r_{ij}, w_{ij})_{j \in \{0,1,2\}}$

Simulated (u, r, w) with $\delta = [0. \ 0.2 \ 0.4]$ and $c = [0. \ 0.4 \ 0.1]$



- **CCPs:** $\Pr(i \text{ chooses } 2) =$ the % of times where $w_{i2} > w_{i1}$ and $w_{i2} > w_{i0}$.
- *Discuss:* Why would it be more correct if $r_{i0} = \infty$? (hint: we have normalized $c_{i0} = 0$)

Model

Consumption utility is $u_{ij} = \delta_{ij} + \varepsilon_{ij}$ where $\varepsilon_{ij} \sim F^0(\cdot)$. Search costs are distributed $c_{ij} \sim F_{ij}^c$, where $F_{ij}^c(c) = F_0^c(c - \mu_{ij})$.

- Given the model, we can define
 - The value of search, $H_{ij}(z) = H_0(z - \delta_{ij})$, with $H_0(z) \equiv \int_{-\infty}^{\infty} \max(\varepsilon - z, 0) dF^0(\varepsilon)$.
 - The reservation value $r_{ij} \equiv \delta_{ij} + H_0^{-1}(c_{ij})$. (The inverse is well-defined so long as F^0 is “well-behaved”, it turns out)
 - The Weitzman index, $w_{ij} \equiv \min\{r_{ij}, u_{ij}\}$, and the final choice, $d_i = \arg \max_{j \in \mathcal{J}} w_{ij}$.
- **MSW** idea: *Boy I wish $w_{ij} \sim T1EV$, then CCPs would be easy...*
- **Idea:** What to assume about F^c , so that $F_{ij}^w(w) = F_0^\varepsilon[w - (\delta_{ij} - \mu_{ij})]$
... because this will imply that $\Pr(i \text{ chooses } j) = \frac{\exp(\delta_{ij} - \mu_{ij})}{\sum_{k \in \mathcal{J}} \exp(\delta_{ik} - \mu_{ik})}$.

- **Reservation utilities** have cdf $r_{ij} \sim F_{ij}^r$, where

$$F_{ij}^r(r) \equiv \Pr(r_{ij} \leq r) = \Pr\left[H_{ij}^{-1}(c_{ij}) \leq r\right] = 1 - F_{ij}^c\left[H_{ij}(r)\right].$$

- **Choice indices** w_{ij} thus have CDF $w_{ij} \sim F_{ij}^w$ where

$$\begin{aligned} F_{ij}^w(w) &= \Pr\left[\min\{r_{ij}, u_{ij}\} \leq w\right] \\ &= 1 - \Pr\left[w < \min\{r_{ij}, u_{ij}\}\right] \\ &= 1 - \Pr\left[w < r_{ij} \text{ and } w < u_{ij}\right] \\ &= 1 - F_{ij}^c\left[H_{ij}(w)\right] \left[1 - F_{ij}^u(w)\right] \\ &= 1 - F_0^c\left[H_0(w - \delta_{ij}) - \mu_{ij}\right] \left[1 - F_0^e(w - \delta_{ij})\right]. \end{aligned}$$

- **Taking stock:** We showed

$$F_{ij}^w(w) = 1 - F_{ij}^c[H_{ij}(w)] \left[1 - F_{ij}^u(w)\right].$$

- **Question:** What should F^c be for $F_{ij}^w(w) = F_0^\varepsilon[w - (\delta_{ij} - \mu_{ij})]$? Let's solve for it:

$$\begin{aligned} F_{ij}^w(w) &= F_0^\varepsilon[w - (\delta_{ij} - \mu_{ij})] \\ 1 - F_{ij}^c[H_{ij}(w)] \left[1 - F_{ij}^u(w)\right] &= F_0^\varepsilon[w - (\delta_{ij} - \mu_{ij})] \\ \Leftrightarrow F_{ij}^c[H_{ij}(w)] \left[1 - F_0^\varepsilon(w - \delta_{ij})\right] &= 1 - F_0^\varepsilon[w - (\delta_{ij} - \mu_{ij})] \\ \Leftrightarrow F_{ij}^c[H_{ij}(w)] &= \frac{1 - F_0^\varepsilon[w - (\delta_{ij} - \mu_{ij})]}{1 - F_0^\varepsilon(w - \delta_{ij})} \end{aligned}$$

- **This** equality holds $\forall w \in \mathbb{R} \Rightarrow$ it also holds for $w = H_{ij}^{-1}(c)$ for every $c \in (0; \infty)$

$$\Rightarrow F_{ij}^c(c) = \frac{1 - F_0^\varepsilon[H_{ij}^{-1}(c) - (\delta_{ij} - \mu_{ij})]}{1 - F_0^\varepsilon[H_{ij}^{-1}(c) - \delta_{ij}]}. \quad \blacksquare$$

Proposition 1 of MSW (2022)

Assume that the search cost distribution is

$$F_{ij}^c(c) = \frac{1 - \exp \{ - \exp [- H_0^{-1}(c) - \mu_{ij}] \}}{1 - \exp \{ - \exp [- H_0^{-1}(c)] \}} \quad (\mu_{ij} \geq 0)$$

Then $w_{ij} \sim \text{T1EV}(\delta_{ij} - \mu_{ij})$, i.e.

$$F_{ij}^w(v) = \exp \{ - \exp \{ - [v - (\delta_{ij} - \mu_{ij})] \} \}.$$

- **Implication:** Final purchase probabilities are in **closed form**

$$\Pr(i \text{ chooses } j) = \Pr(w_{ij} \geq w_{ik} \forall k \neq j) = \frac{\exp(\delta_{ij} - \mu_{ij})}{\sum_{k \in \mathcal{J}} \exp(\delta_{ik} - \mu_{ik})}.$$

Choice Probabilities

$$\Pr(i \text{ chooses } j) = \frac{\exp(\delta_{ij} - \mu_{ij})}{\sum_{k \in \mathcal{J}} \exp(\delta_{ik} - \mu_{ik})}.$$

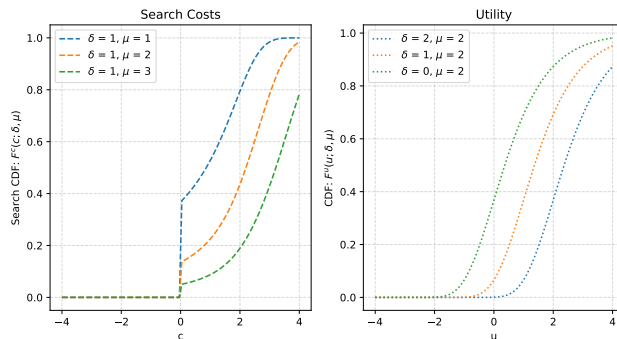
- **Recall:** δ_{ij} shift (mean) *utility*, while μ_{ij} shifts (mean) *search costs*.

Identification problem

Utility and search cost shifters are not separately identified using data on final purchase alone.

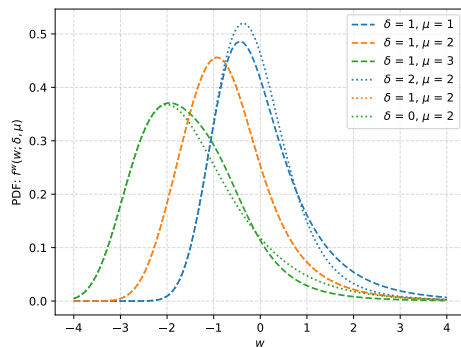
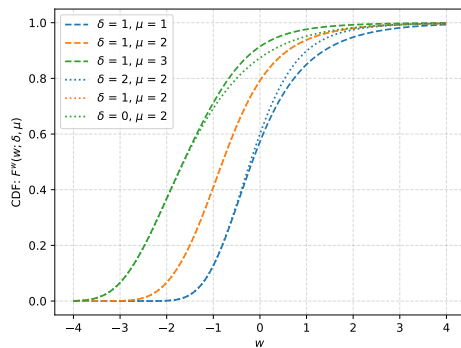
- **Implication 1:** The model is just vanilla logit, which we can simply estimate.
- **Implication 2:** Mean utility and search costs enter linearly – so $\delta_{ij} \uparrow$ and $\mu_{ij} \downarrow$ cancel out.

Changing Parameters



- **Utility:** δ_{ij} just shifts the distribution of u_{ij} horizontally.
- **Search costs:** μ_{ij} induces a *non-linear* shift in c_{ij} . But *intuitively* think of it as shifting the mean.

Visualizing the CDF of $w_{ij} \equiv \min\{r_{ij}, u_{ij}\}$



Note: Here, we look at the distribution of w_{ij} , changing (δ, μ) .

Distributions are changed, but *choice probabilities* are not (because all options are symmetric)

Probability of not searching?

- **Question:** What is the probability of not searching (beyond the first search, assuming $c_{i0} = 0$)?
- **This occurs** if the outside utility was below all inside reservation values

$$\begin{aligned}\Pr(\text{no search}) &= \Pr(r_{ij} \leq u_{i0} \forall j = 1, \dots, J) \\ &= \mathbb{E} \left[\prod_{j=1}^J \Pr(r_{ij} \leq u_{i0} | u_{i0}) | u_{i0} \right] \\ &= \int_{-\infty}^{\infty} \left[\prod_{j=1}^J F_{ij}^r(u_{i0}) \right] f_0^u(u_{i0}) du_{i0}.\end{aligned}$$

and recall that $F_{ij}^r(z) = \Pr(r_{ij} \leq z) = \Pr[c_{ij} \leq H_{ij}(z)] = F_{ij}^c[H_{ij}(z)]$.

```
1 from scipy.integrate import quad
2 f_pdf_u = lambda u : stats.gumbel_r.pdf(u)
3 Fr = lambda u : F_c(H(u, j), j) # assuming we have F_c and H defined
4 f_prod = lambda u : Fr(u, j=1) * Fr(u, j=2) * Fr(u, j=3) # ex. for J=3
5 f_integrand = lambda u : f_prod(u) * f_pdf_u(u)
6 Pr_no_search = quad(f_integrand, -np.inf, np.inf)[0]
```

- **It's always true** that $F_{ij}^w(w) = 1 - F_{ij}^c[H_{ij}(w)] [1 - F_{ij}^u(w)]$, and

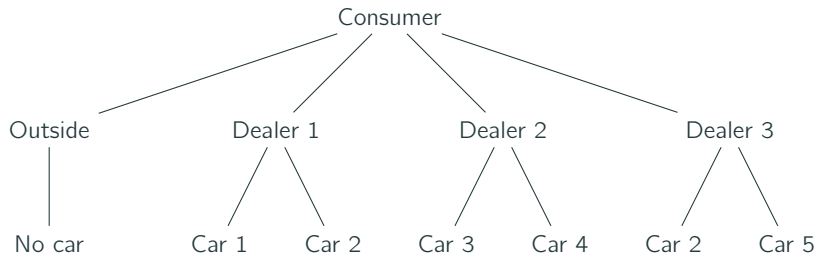
$$\Pr(i \text{ chooses } j) = \Pr(w_{ij} \geq w_{ik} \forall k \neq j) = \int_{-\infty}^{\infty} [\prod_{k \neq j} F_{ik}^w(z)] dF_{ij}^w(z)$$

- **If we assume** that F_{ij}^c takes the “logit-preserving” form, it simplifies to

$$\Pr(i \text{ chooses } j) = \frac{\exp(\delta_{ij} - \mu_{ij})}{1 + \sum_{k=1}^J \exp(\delta_{ik} - \mu_{ik})}.$$

- **The qualitative** effects of search cost shifters are the same:
 - Better ($\delta_{ij} \uparrow$) or easier to find ($\mu_{ij} \downarrow$) products $\Rightarrow$ higher market share.
- **If we have data on search behavior**, we may be able to avoid the identification issues stemming from the linearity of $\delta_{ij} - \mu_{ij}$...

Car Search



1. Visit dealerships (sequential search problem),
2. Choose the best car among all those you have seen.

- **Until now:** Consumers search for *products* directly.
- **MSW:** In Moraga-Gonzales, Sandor & Wildenbeest (2022), they search *dealers*.
- **Market setting:** New car purchases (as in BLP)

Final Consumption Utility

- Final consumption utility is ala BLP

$$u_{ij} = \delta_{ij} + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \text{IID T1EV},$$

$$\delta_{ij} \equiv \mathbf{x}'_j \boldsymbol{\beta}_i + \alpha_i p_j + \xi_j$$

- **Information structure:** δ_{ij} known prior to search; ε_{ij} (“match quality”) only learnt after search.
- **Random Coefficients:** $\alpha_i \equiv \tilde{\alpha}/y_i$, and $\boldsymbol{\beta}_i \sim \mathcal{N}(\bar{\boldsymbol{\beta}} + \Pi D_i, \Sigma)$, where D_i are demographics.
- **Econometrician observes** $(\mathbf{x}_j, p_j)$ and the distribution of (y_i, D_i) .
- **Price expectations?** Consumers have correct conjectures about equilibrium prices during search, but may encounter a deviation after search. Like in Wolinsky/Anderson-Renault (WAR)

Sequential search

- **Distinction:** A *firm*, f , is a “dealership”, which carries a range of products, $\mathcal{J}_f \subset \mathcal{J} \equiv \{1, \dots, J\}$.
- **Search choice:** Consumer i must choose which firm (dealership) to visit, $f \in \{1, \dots, F\}$.
 - Once at the dealership, she immediately learns ε_{ij} for all $j \in \mathcal{J}_f$, and picks her preferred car, gaining

$$\text{utility from visiting firm } f \equiv \delta_{if} \equiv \max_{j \in \mathcal{J}_f} u_{ij}$$

Recall Lemma 2.2.2 (max-stability of T1EV)

Suppose $u_{ij} \sim \text{T1EV}(\delta_{ij}) \forall j \in \mathcal{J}_f$, and $u_i^* \equiv \max_{j \in \mathcal{J}_f} (u_{ij})$. Then

$$u_i^* \sim \text{T1EV}(\delta_i^*) \quad \text{with } \delta_i^* \equiv \log \sum_{k \in \mathcal{J}_f} \exp(\delta_{ik}).$$

- **Conclusion:** the choice over which firm to visit is a logit with utility

$$\begin{aligned} v_{if} &= \delta_{if}^{\text{firm}} + \epsilon_{if}, \quad \epsilon_{if} \sim \text{IID T1EV}(0) \\ \delta_{if}^{\text{firm}} &\equiv \log \sum_{j \in \mathcal{J}_f} \exp(\delta_{ij}). \end{aligned}$$

- **In general:** we can use the Law of Iterated Expectations to write

$$\Pr(i \text{ chooses } j) = \sum_{f=1}^F \Pr(i \text{ chooses } j | \text{visiting } f) \times \Pr(\text{visit } f)$$

- If dealerships are **exclusive** (so $\mathcal{J}_f \cap \mathcal{J}_g = \emptyset$), the sum “disappears”.
- **Conditional on search**, the choice within firm i is

$$\Pr(i \text{ chooses } j | \text{searching } f) = \frac{\exp(\delta_{ij})}{\sum_{k \in \mathcal{J}_f} \exp(\delta_{ik})} = \frac{\exp(\delta_{ij})}{\exp(\delta_{if}^{\text{firm}})}.$$

- **No search?** If $c_{ij} = 0 \forall j$, then “ $r_{ij} = \infty$ ”, so $w_{ij} = u_{ij}$.
 - $\Rightarrow$ thus, we just have a regular logit: $\Pr(i \text{ chooses } j) = \frac{\exp(\delta_{ij})}{1 + \sum_k \exp(\delta_{ik})}$.
 - $\Rightarrow$ our search framework **nests** the usual BLP model.

MSW (2022) Model

- Consumer i is faced with $f \in \{0, 1, \dots, F\}$ firms,
- Choosing firm f yields utility $v_{if} \equiv \delta_{if}^{\text{firm}} + \epsilon_{if}$, where $\epsilon_{if} \sim \text{T1EV}$.
 - Choosing firm f is equivalent to choosing $\max_{j \in \mathcal{J}_f} u_{ij}$.
- Each firm searched incurs search cost $c_{if} \sim F_{if}^c(\cdot)$

- **Search cost** distribution can be simplified as $F_{if}^c(c) = F_0^c(c - \mu_{if})$
 - Equivalent to assuming $c_{if} = \mu_{if} + v_{if}$, with $v_{if} \sim F_0^c$.
- **Proposition 1:** If F_{if}^c takes a specific form, then s_{ij} simplifies:

$$F_{if}^c(c) = \frac{1 - F_0^\epsilon[\mu_{if} + H_0^{-1}(c)]}{1 - F_0^\epsilon[H_0^{-1}(c)]} \implies \Pr(i \text{ chooses } j) = \frac{\exp(\delta_{ij} - \mu_{if_j})}{1 + \sum_{k=1}^J \exp(\delta_{ik} - \mu_{if_k})}.$$

- **Aggregation** is just by “summing” (or integrating) over consumers:

$$s_j = \sum_i \Pr(i \text{ chooses } j) = \int \frac{\exp(\delta_{ij} - \mu_{if_j})}{1 + \sum_{k=1}^J \exp(\delta_{ik} - \mu_{if_k})} dF(\boldsymbol{\delta}_i, \boldsymbol{\mu}_i).$$

Market share

- **Supply:** If $\delta_{ij} = \alpha_j p_{ij} + \dots$, then $s_j = s_j(p_j)$, and firms solve the usual

$$\max_{p_j} (p_j - mc_j) s_j(p_j, \mathbf{p}_{-j}).$$

- **Endogeneity?** The same price endogeneity concerns persist here.

- **Estimation?** Either using

- Market level GMM: $\min \|\xi(\theta)\|$, possibly $\|\xi(\theta) \times Z\|$
- Market-level with search: $\left\| \begin{pmatrix} \xi(\theta) \\ \Pr(\text{never search}|\theta) - N^{-1} \sum_i \mathbf{1}\{\text{never search}\}_i \end{pmatrix} \right\|$
- Micro-level, using only purchase information: $\ell_i(\theta) = \log \Pr(i \text{ chooses } d_i)$
- Micro-level, using search data: $\ell_i(\theta) = \log \Pr(i \text{ chooses } d_i \text{ and searches } \mathcal{J}_i)$.

Ursu 2018

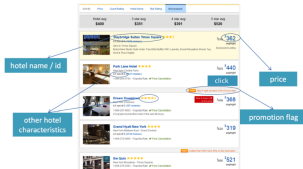
Figure 1: Information on the Data Observed



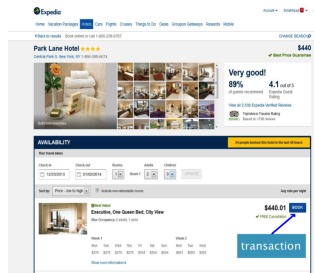
The image shows the Expedia search interface. At the top, there are input fields for 'visitor country, region', 'site name', and 'time of search'. Below these are navigation links: 'Home', 'Vacation Packages', 'Hotels', 'Cars', 'Flights', 'Cruises', and 'Things to Do'. The main section is titled 'PLAN YOUR TRIP ON EXPEDIA' and includes a 'Flight + Hotel' selection box with a '140,000 HOTELS WORLDWIDE' badge. Below this is a 'Hotel' section with a 'Find hotels near:' dropdown and a 'destination' input field. There are also fields for 'length of stay', 'booking window', '# adults', '# children', and '# rooms'. A 'SEARCH FOR HOTELS' button is at the bottom right.

Expedia Inc.

Search Query



Search Impression: Information Before the Click



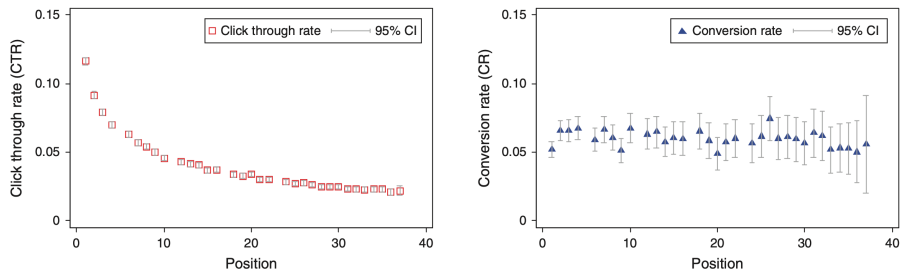
Hotel's Page: Information After the Click

- **Empirical context:** Expedia – online platform for booking hotels.

- **Expedia:** Online platform for booking hotels.
- **Dataset** on search queries, clicks (“search”), hotel bookings (“choice/purchase”), and hotel characteristics (price, rating, etc.)
- **Positioning:** Either *a*) according to Expedia’s algorithm, or *b*) **randomized** (!!!!)
 - **Randomization:** There *must* be a great way of exploiting this.
- **Ursu’s** contributions are *i*) providing empirical insights, and *ii*) testing specific hypotheses from a search model.

Click-Through-Rates (CTR) and Conversion Rates (CR)

Figure 1. (Color online) The Effect of Position on Click-Through Rate and Conversion Rate: Random Ranking



- **Distinction:** $\text{CTR} = \Pr(\text{search})$, and $\text{CR} = \Pr(\text{purchase}|\text{search})$.
- **Takeaway:** $\Pr(\text{purchase}|\text{search})$ is independent of position!!!

Discuss

Explain why *i*) CTR is declining in position, and *ii*) CR is constant.

Randomization changes CR, but not CTR

Figure 1. (Color online) The Effect of Position on Click-Through Rate and Conversion Rate: Random Ranking

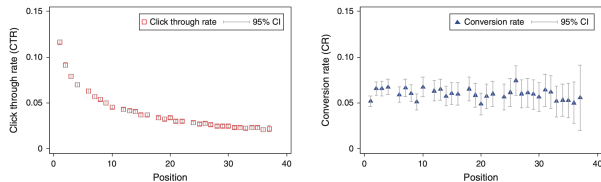
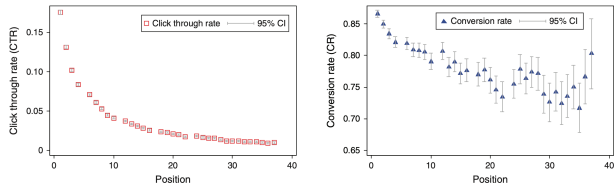


Figure 2. (Color online) The Effect of Position on Click-Through Rate and Conversion Rate: Expedia Ranking



Note. The scale of the y-axis in the two figures is different.

Discuss

1. Why is effect of position on $\Pr(\text{search})$ the same in the random and non-random samples?
2. Why is it different for $\Pr(\text{purchase}|\text{search})$ (CTR)?

Empirical Model from Ursu (2018)

- **Utility** of purchase is

$$u_{ij} = v_{ij} + \epsilon_{ij}, \quad \epsilon_{ij} \sim \mathcal{N}(0, \sigma_j^2),$$
$$v_{ij} = \mathbf{x}'_{ij}\boldsymbol{\beta}.$$

- **Search costs** are *deterministic* (no error term)

$$c_{ij} = \exp(k + \gamma P_{ij}), \quad P_{ij} = \text{position in the list} \in \{1, \dots, J\}$$

- **Functional form:** Ursu assumes *Gaussian* utility,

$$u_{ij}^{\text{Ursu}} \sim \mathcal{N}(\mathbf{x}'_{ij}\boldsymbol{\beta}, \sigma_j^2), \quad u_{ij}^{\text{MSW}} \sim \text{T1EV}(\mathbf{x}'_{ij}\boldsymbol{\beta}).$$

- *Note:* extra parameters, σ_j ! **But:** scale identification is needed $\Rightarrow$ she sets $\sigma_j = 1 \ \forall j$.
- **Cool:** For normal utility, $H_{ij}(z)$ is available in closed form!

$$H_0(z) = \phi(z) - z [1 - \Phi(z)], \quad H_{ij}(z) = \sigma_j H_0\left(\frac{z - v_{ij}}{\sigma_j}\right).$$